

PROJECT AUTOMATION WORKFLOW

Standard Operating Procedure | Kingdom Structural LLC | v2.1.11 May 2026

1. What This Document Covers

This procedure explains how to take a new project from inquiry to signed contract using the automation pipeline. The pipeline runs seven jobs that together: create a Proposal Brief, set up a Fee Schedule with 6 standard service rows, create the OneDrive folder with 13 subfolders, generate a Fee Analysis Memo PDF for partner review, notify the assigned engineer, and render the final contract Word document once the brief is approved. Each step runs only when the previous step's requirements are met.

If something does not fire as expected, jump to Section 11 (Troubleshooting). Most issues come from a missing required field on the Project page in Notion.

2. The Automation Pipeline at a Glance

Seven automated jobs run in sequence. The team interacts with the pipeline at three points: project creation, brief filling, and brief approval. The rest is automatic.

1. Admin creates the Project record in Notion, fills required fields, and checks Intake Complete.
2. Job A0 creates the Proposal Brief automatically.
3. Job A02 creates the Fee Schedule child database with 6 standard service rows.
4. Job A creates the OneDrive folder structure (13 subfolders) and stamps the SharePoint URL on the Project.
5. Job F generates the Fee Analysis Memo PDF and drops it in the project folder.
6. Job C notifies the assigned engineer that the project is ready for them.
7. Engineer fills out the Brief, sets Status to Ready for Admin Review (fires Job D, which notifies admin).
8. Admin reviews, makes corrections, and sets Brief Status to Approved.
9. Job B renders the final Contract document into the project's Contracts folder.

3. Before You Start

These are one-time setup items per user.

- Notion access with edit rights to: Projects, Proposal Brief, Clients, Contacts, Automation Log databases.
- OneDrive desktop client signed in to your Kingdom Structural account, with the _Projects folder visible in File Explorer.
- Right-click the _Projects folder → Always keep on this device, so the automation can read folder paths reliably.
- Sweep script running on Michael's laptop via Windows Task Scheduler. If it is not running, the cascade will not fire even if your fields are perfect — talk to admin team.

4. Creating a New Project — Step by Step

This is the only manual part. Once you complete this section correctly, the automation handles everything else.

Step 1: Open the Projects database

In Notion, navigate to The Plate, then click on the Projects database. Click + New to create a project page using the standard template (the page opens with an Intake Checklist callout at the top).

Step 2: Fill in the required fields

Every field below must be populated before the automation will fire. The Intake Status formula at the bottom of the property list shows you which one is missing if you are not sure.

Field	Type	What to put here
Project Name	Title	Must start with the 8-digit KS job number (year + sequence + client number). Example: 26124149 Monte Vista Residential Remodel.
Client	Relation	Pick from the Clients database. Create a new Client first if needed (with Name, Address, Number).
Project Contact	Relation	Architect, owner, GC, etc. — pick from Contacts database. Create new Contact if needed.
ENGINEER	People	Assigned KS engineer. Gets the notification when intake completes.
Admin	People	Assigned KS admin. Usually the person creating the record.
City / State	Text	Project city and state. Use full state name or abbreviation.
Project Street	Text	Full street address of the project site.
Project Type	Select	Pick from the dropdown — drives fee memo comp matching. See Section

		5 for the 16 categories.
Engineering Status	Status	Set to 'Proposal Requested'. Required for cascade to fire.

Step 3: Verify Intake Status, then check Intake Complete

Scroll to the Intake Status property. When everything is filled correctly it shows '✓ Ready for Proposal Requested'. If you see a warning like '⚠ Project Type missing (required for fee memo)', fix the named field — formula updates immediately. Once Ready, check the Intake Complete checkbox at the top of the property list. This triggers the cascade.

After checking Intake Complete, wait 30–60 seconds, then refresh. You should see the Brief relation populated, Status flip to 'Proposal Brief Pending', and a SharePoint Folder URL stamped in.

5. The 16 Project Type Categories

Project Type drives the Fee Analysis Memo's comp matching. Pick the closest fit:

- Residential – Remodel / Addition: Existing home remodels, expansions, additions; ADUs attached to existing homes.
- Residential – Single-Family: New single-family home (ground up).
- Residential – Multi-Family: Apartments, duplexes/triplexes, hotel/motel, condo buildings.
- Religious: Churches, religious community centers, parish halls.
- Restaurant: Restaurant TIs or new restaurant builds.
- Storage: Self-storage, warehouses, industrial racking.
- Medical: Dental offices, MRI suites, urgent care, healthcare facilities.
- Office: Office TIs or office buildings.
- Education: Schools, training facilities, educational TIs.
- Commercial – TI: General commercial tenant improvement (retail, services, mixed-use).
- Industrial: Manufacturing, processing, mechanical/HVAC contractor facilities.
- Canopy: Standalone canopies, fuel canopies, drive-thru covers.
- Specialty Structures: Custom one-off structural elements (gantries, trellises, towers, large retaining walls).
- Existing Building / Assessment: Inspection reports, structural observations, peer review, repair scope without redesign.
- PEMB: Pre-engineered metal buildings (foundation and anchor design).
- Other: Use only when no category fits — flag with the partner.

6. What the Automation Does After Intake Complete

Job A0 — Proposal Brief created (~5 sec)

New Brief page in the Proposal Brief database, linked to your Project, pre-populated with the standard scope-of-services checkbox template. Click the Proposal Brief relation on your Project to jump there.

Job A02 — Fee Schedule child database (~5 sec)

Inside the new Brief, a child database called 'Fee Schedule' is created with six standard rows: Schematic Design, Design Development, Construction Documents, Construction Administration, Special Structural Inspections, Engineering Site Visits.

Job A — OneDrive folder created (10–30 sec)

Folder created at: OneDrive – Kingdom Structural LLC_Projects\YEAR\JOBNUMBER ProjectName - City, State\. Inside, 13 subfolders are created automatically: Calculations, City Comments, Construction Administration, Contracts, Correspondence, Drafting, Drawings, Geotech Report, Notes, QAQC, Redlines, Sent to Client, Site Visit. On the Project page in Notion: Folder checkbox flips to ✓, SharePoint Folder URL is stamped (clickable), Status moves to 'Proposal Brief Pending'.

Job F — Fee Analysis Memo PDF (15–30 sec)

A one-page Fee Analysis Memo PDF is generated and dropped at the top level of the project folder. Filename: '{8-digit job number} Fee Analysis Memo.pdf'. Both .docx editable source and .pdf are saved. The memo is partner-facing internal-only — covers project metadata, top 5 comparable wins (filtered by Project Type), loss signals at similar fees, fee distribution stats (median, P75, max, win rate), recommended fee tiers (Conservative/Market/Stretch with \$/SF), phase distribution (Design vs CA per tier), industry benchmarks, and a Partner Decision section with tier-selection checkboxes.

Important: the fee memo is internal-only. It contains historical data on past wins and losses. Do not share with clients.

Job C — Engineer notified (~1 min)

Assigned engineer gets a Notion comment on the Project page tagging them with a link to the new Brief and a note that everything is ready.

7. The Engineer's Role — Filling Out the Brief

Once notified, the engineer fills in project-specific scope and fees. Every choice flows into the final rendered contract.

10. Open the Brief via the Proposal Brief relation on the Project page.
11. Fill in metadata: ICC Code Year, Jurisdiction, Approx. Structural SF, Status (move to In Progress while working).

12. In the Brief body, check applicable scope-of-services bullets across each section: Custom Residence, Residential Remodel, Mechanical T.I., PEMB, Site Designs, Geotechnical, Drafting by Others, Submittals, Self-Cert, Engineering Site Visits, Construction Admin, Special Structural Inspection. Each checked bullet renders into the contract. Edit placeholder x,xxx values to actual square footage.
13. Pick exactly one Reimbursables Treatment: Standard (billed at cost) / Digital Only (no reimbursables) / Included in Fee.
14. Check Basic Services bullets (Engineering Site Visits, Structural Calculations, AutoCAD/Revit drawings, CA, etc.) — they flow into the contract's basic-services paragraph.
15. Open the Fee Schedule child database. For each applicable row: set Type (Fixed Fee / Hourly / Not to Exceed) and Amount (>0). Rows where Amount is 0 or empty are excluded automatically via the Include formula.
16. Set Brief Status to 'Ready for Admin Review'. Job D fires and notifies admin team.

Use the Fee Analysis Memo PDF in the project folder as your fee reference. Its tier table and phase distribution give suggested per-row dollar amounts.

8. Admin's Role — Reviewing and Approving

Admin/partner reviews Brief, references Fee Analysis Memo PDF for pricing, makes corrections, and approves for contract render.

17. Open the Brief from the Job D notification.
18. Verify scope bullets reflect actual work. Cross-reference Fee Schedule against Fee Analysis Memo PDF. Adjust amounts if pricing changes. Use the memo's Partner Decision section to record tier selection and reasoning.
19. Set Brief Status to 'Approved'. Job B fires within 60 seconds.

9. Job B — Contract Renders Automatically

Job B picks up Briefs with Status=Approved and empty Rendered At. It pulls Project, Brief, Client, Contact, Fee Schedule, and engineer info; loads the contract template; substitutes placeholders; builds the Scope of Services from checked bullets; builds Basic Services paragraph; builds Fee Schedule table from rows where Amount>0; saves the contract Word document to the project's Contracts subfolder; stamps Brief Status to 'Rendered' and Rendered At with today's date; stamps Project.Status to 'Contract Rendered'; logs to Automation Log.

Output path: OneDrive_Projects\YEAR\JOBNUMBER ProjectName\Contracts\{job#} Contract.docx. If re-rendered (Brief edited and re-approved), new file gets v2/v3 suffix — original never overwritten.

10. Final Project Folder Contents

After all jobs run, the project folder contains: {job#} Fee Analysis Memo.docx and .pdf at the project root (internal review documents); 13 subfolders — Calculations, City Comments, Construction Administration, Contracts (with {job#} Contract.docx for the client), Correspondence, Drafting, Drawings, Geotech Report, Notes, QAQC, Redlines, Sent to Client, Site Visit.

The Fee Analysis Memo lives at the project root (not inside Contracts) because it is internal-only. The Contracts subfolder is reserved for client deliverables.

11. Troubleshooting

Intake Complete will not trigger anything

- Check Intake Status formula on the Project. If it says 'Missing X', fill in field X and re-check.
- If it says 'Ready' but nothing fires, the sweep script might be paused — contact admin team.

OneDrive folder did not appear

- Wait 60 seconds — sweep runs on a schedule, not instantly.
- Check Project.Folder checkbox — if ticked, folder was created (check OneDrive sync). If not ticked, check Automation Log for a Job A row tagged Failed.
- Right-click _Projects folder → Always keep on this device — forces OneDrive to fully download.

Fee Analysis Memo did not generate

- Check Project Type is set on the Project page — Job F requires it.
- Check the Folder checkbox is ticked.
- Check Automation Log for a Job F row. If 'PDF conversion failed', Microsoft Word needs to be installed OR the docx2pdf Python package needs installing. If 'folder not yet on disk', OneDrive has not synced yet — wait and retry.

Engineer did not get notified

- Check Project.ENGINEER property — is a person assigned?
- Check Notion notifications — sometimes the comment tag goes to a different inbox.
- Check Automation Log for the Job C row — Success or Failed?

Contract did not render

- Check Brief Status — must be 'Approved'. Check Brief.Rendered At — if it has a date, the contract is already in the Contracts folder.
- Common Brief data issues: missing Reimbursables Treatment, no Fee Schedule rows with non-zero amounts, missing City/State.

12. Maintenance

- Weekly: review the Automation Log database in Notion, filter to past 7 days, check for Failed rows. Each failure has an Error column explaining what went wrong.
- Monthly: refresh the comp database. Make sure Master-Business Plan.xlsm has recent project fees, then run 'python import_excel_comps.py --force' from the _automation folder. Rebuilds comps_cache.json with the latest data.
- As needed: refresh Excel Client Insights sheet — close Excel, run 'python refresh_client_insights_sheet.py', reopen and click the Client Insights tab.
- As needed: bulk-classify Project Types — edit project_type_assignments.csv with proposed types and run 'python bulk_assign_project_type.py --apply'.

13. Keeping This Document Current

This SOP is a living document. The automation code and this procedure must stay in sync — if they drift, partners and admins lose trust in both. The following changes always require an SOP update in the same workweek as the change:

- New job added or existing job removed (e.g., a new Job G or retiring Job F) — update Sections 2, 6, and Troubleshooting.
- Job trigger conditions modified (filter changes in sweep.py, new gates added to a workflow) — update the relevant job's section and Troubleshooting.
- Notion property added, removed, or renamed on Projects, Brief, Clients, or Contacts — update the required-fields table in Section 4 and any field references throughout.
- Intake Status formula changed (new field check added, validation message wording changed) — update Section 4 and Troubleshooting.
- Project Type select options added or removed — update Section 5.
- Folder structure changes (new subfolder added, naming convention modified) — update Sections 6 and 10.
- Contract template placeholder changes — update Section 9 and any references to fields the template reads.

Owner: whoever modifies the automation code, Notion schema, or workflow definitions is responsible for updating this SOP in the same commit or pull request as the change. Note an 'SOP update needed' item if you cannot do it in the same commit, and don't merge code that diverges from the SOP without a tracking task.

Source of truth: this SOP lives in the same git repository as the automation code (in the SOP folder). Changes flow through standard review. Distribute updated PDF copies to the team after every revision.